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Identifying the uneven distribution of health and education services in China using open geospatial data

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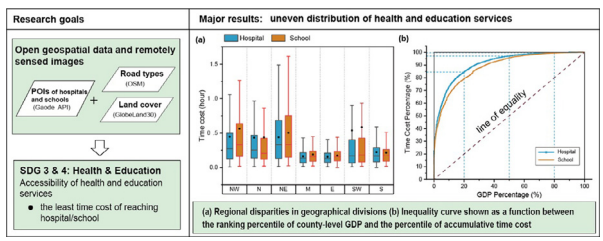
HIGHLIGHTS

- The accessibility was measured using POI, OSM, land cover, and WorldPop data.
- The accessibility was measured in terms of travel time to the nearest service.
- The average time costs of hospital and school are 20 and 22 min, respectively.
- Accessibility is high in eastern China and low in western China.
- Low-GDP counties require high mean time cost to reach hospital and school.

GRAPHICAL ABSTRACT



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ABSTRACT

Growing attention has been directed to the use of satellite imagery and open geospatial data to understand large-scale sustainable development outcomes. Health and education are critical domains of the Unites Nations' Sustainable Development Goals (SDGs), yet existing research on the accessibility of corresponding services focused mainly on detailed but small-scale studies. This means that such studies lack accessibility metrics for large-scale quantitative evaluations. To address this deficiency, we evaluated the accessibility of health and education services in mainland China in 2021 using point-of-interest data, OpenStreetMap road data, land cover data, and WorldPop spatial demographic data. The accessibility metrics used were the least time costs of reaching hospital and school services and population coverage with a time cost of less than 1 h. On the basis of the road network and land cover information, the overall average time costs of reaching hospital and school were 20 and 22 min, respectively. In terms of population coverage, 94.7% and 92.5% of the population in China has a time cost of less than 1 h in obtaining hospital and school services, respectively. Counties with low accessibility to hospitals and schools were highly coupled with poor areas and ecological function regions, with the time cost incurred in these areas being more than twice that experienced in non-poor and non-ecological areas. Furthermore, the cumulative time cost incurred by the bottom 20% of counties (by GDP) from access to hospital and school services reached approximately 80% of the national total. Low-GDP counties were compelled to suffer disproportionately increased time costs to acquire health and education services compared with high-GDP counties. The accessibility metrics proposed in this study are highly related to SDGs 3 and 4, and they can serve as auxiliary data that can be used to enhance the evaluation of SDG outcomes. The analysis of the uneven distribution of health and education services in China can help identify areas with backward public services and may contribute to targeted and efficient policy interventions.

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1. Introduction

The United Nations' Sustainable Development Goals (SDGs) paint a shared blueprint for the world to achieve peace and prosperity over the next several years (United Nations, 2015). Among them, SDGs 3 and 4 focus on the health and education dimensions of human well-being, respectively. These dimensions are also particularly emphasized both in China's poverty reduction goals (SCIO, 2021) and in the international measurement of multi-dimensional poverty (Alkire and Foster, 2011). Narrowing the inequality of access to health and education services is critical to human well-being and the SDGs. This requirement has prompted a demand to measure the accessibility of such services, track progress in goal achievement, and establish efficient policy interventions.

Accessibility is a widely employed concept in transportation, urban planning, and geography; it has no definitive definition, but it typically refers to the ease or cost encountered by an individual or an area in the process of reaching certain services or opportunities (Hansen, 1959; Weiss et al., 2018; Stępniak et al., 2019; Handy, 2020). Many studies have measured the accessibility of different public services, such as healthcare facilities (Ghorbanzadeh et al., 2021), supermarkets (Niedzielski and Kucharski, 2019), parks and gardens (Shoari et al., 2020), job opportunities (Deboosere and El-Geneidy, 2018), and even COVID-19 testing sites (Tao et al., 2020). Numerous approaches have been used to understand and measure accessibility, but we found two major categories of methods: individual-based and location-based accessibility measurements (Geurs and van Wee, 2004; Neutens, 2015). Individual-based metrics measure accessibility from the viewpoint of an individual's demands and characteristics; they are aimed at representing individual activity spaces under various space-time constraints (Chen et al., 2019). Individual-based accessibility is difficult to operationalize for large-scale measurement because it entails highly time-consuming individual-level data collection and processing. Location-based metrics ascertain the accessibility of spatially distributed services or facilities. These are usually classified into three types, namely, proximity measures (travel distance or time to the nearest destination), accumulative opportunity measures (contour measure wherein there are more than two destinations), and potential accessibility measures (gravity-based measures) (Geurs and van Wee, 2004). Regardless of the measurement method, there are four key components of accessibility: transportation systems, land use, temporal factors, and individual characteristics (Stępniak et al., 2019). Existing accessibility metrics focus primarily on one or several of the four components of accessibility (Geurs et al., 2015). The contradiction is that the more detailed the information contained in a metric, the higher the calculation cost, which translates to challenging large-scale quantitative evaluations. In this study, we conducted a large-scale evaluation with the use of a proximity accessibility measure and a consideration of land cover and transportation systems.

With regard to large-scale accessibility metrics, voluminous satellite imagery and open geospatial data have provided new perspectives and technical means through which to measure accessibility (Fu, 2020; Fu et al., 2021; Guo et al., 2021). Weiss et al. (2018) created a 2015 global accessibility map that estimates the travel time required to the nearest urban center using OpenStreetMap (OSM), Google roads datasets, and the Global Human Settlement Grid of high-density land cover. The satellite imagery and geospatial data could provide information on transportation systems, land use and land cover, and population distribution during a large-scale quantitative evaluation of accessibility (Ge et al., 2022). Volunteered geographic information (VGI) is a way of obtaining and applying geographic information driven by the Internet Web 2.0 environment (Goodchild, 2007); this generally refers to geographic information volunteered and created by many non-professional users. The OSM database is one of the most widely used VGI application platforms that provide a wide range of features (i.e., road networks, buildings, and land cover) (Vargas-Munoz et al., 2021). Point-of-interest (POI) data represent real geographical entities and can be obtained from OSM or various map application programming interfaces. POI data are geographical entities closely related to human life, such as a school, bank, hospital, supermarket, and airport, and an entity point typically includes spatial information, such as longitude and latitude, and attribute information, such as category and name (Gao et al., 2017). Some open geospatial demographic datasets are also available, such as WorldPop spatial demographic (WorldPop, 2018) and Global Human Settlement Layer (https://ghsl.jrc.ec.europa.eu/) datasets. These offerings motivated us to deliberate on whether large-scale accessibility metrics can be generated using remote sensing images and multi-source geospatial data.

In this study, we used a satellite imagery-based data product and multi-source geospatial data to evaluate the accessibility of health and education services in China, as measured by the time cost to reach the nearest service. The open and transparent datasets used in this study were POI, OSM road network data, a GlobeLand30 land cover dataset, and WorldPop spatial demographical data. The POI comprised locations of health and education services. The time cost calculation was based on a cost surface, which entails information on road networks, travel speeds, and land cover. This study presents three major contributions: (1) proposing two pixel-level accessibility metrics that enable the evaluation of access to hospitals and schools using open geospatial data; (2) generating two datasets of the time costs incurred in accessing hospital and school services, with a 30 m spatial resolution that covers the entirety of mainland China; and (3) quantifying regional disparities and inequalities in the distribution of health and education services in China.

2. Materials and methods

2.1. Study area

The study area covers the entire area of mainland China, including 2,849 counties in total. In this study, to analyze the disparities and inequalities in the distribution of health and education services, we focused on three groups of areas: counties in different geographical divisions (Fig. 1(a)), poor counties (Fig. 1(b)), and ecological function counties (Fig. 1(b)). Here, we provide a brief introduction to China's poor counties and ecological function counties. The 832 poor counties were identified by the Chinese government at different stages of social and economic development. In 2001, the *Outline for Poverty Reduction and Development of China's Rural Areas (2001–2010)* identified 592 key counties based on the per capita net income of rural households (SCIO, 2011). Then, in 2011, the *Outline for Development Oriented Poverty Reduction for China's Rural Areas (2011–2020)* identified 14 continuous poverty-stricken areas, which include 680 counties as the battleground of a new round of poverty reduction (Liu et al., 2018). The 680 poor counties were determined mainly according to the economic metrics, with consideration of their geographical conditions, traditional industries, and culture and customs. There are 440 overlapping counties between the impoverished counties identified in the two periods. Then, combining the counties identified in the two periods, 832 impoverished counties were included in this study, as shown in Fig. 1(b). In the *National Main Functional Area Plan* issued in 2010, the Chinese government established 25 key national ecological functional areas to protect ecological security (The State Council of the People's Republic of China (SC), 2010). At the end of 2016, the key national ecological functional areas included 676 ecological function counties in total (SC, 2016), as shown in Fig. 1(b). These counties undertake important ecological protection and restoration functions such as water and soil conservation, windbreak and sand fixation, and biodiversity protection. From Fig.1(b), the poor counties are highly overlapped with the ecological function counties. The population density and gross domestic product (GDP) of each county in 2020 are mapped in Fig.1(c) and (d), respectively. It can be observed that the population density and the GDP are both high in the east and low in the west of China.

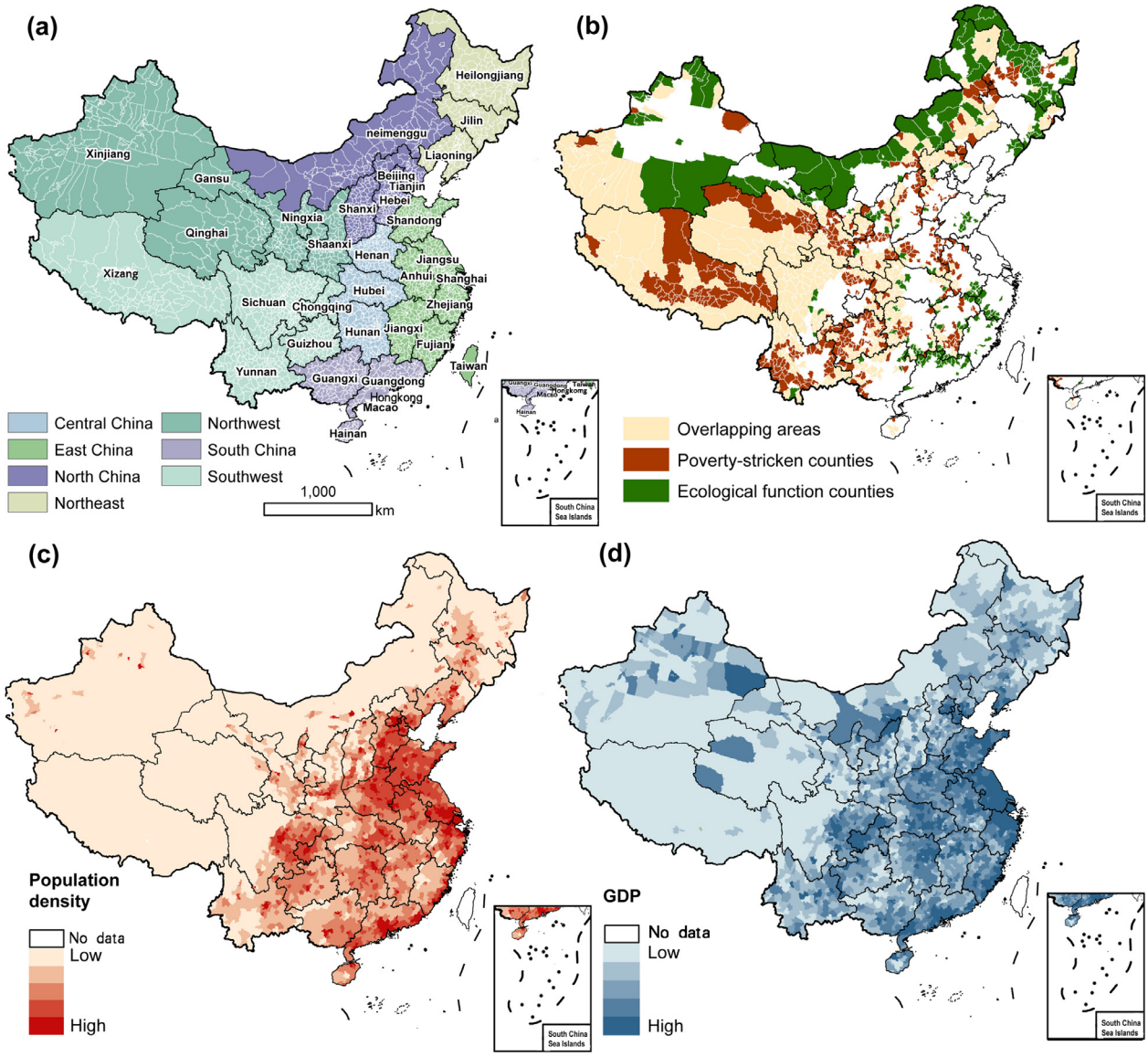

Fig. 1. Study area. (a) The seven geographical divisions of China; (b) distribution of impoverished counties and ecological function counties, the overlapping areas indicate the overlap counties between poor counties and ecological function counties; (c) the population density of each county in 2020; (d) gross domestic product of each county in 2020.

Table 1
Data sources and descriptions.

Data	Sources	Spatial resolution	Year
POIs of school and hospital	Gaode application programming Interface (lbs.amap.com)	–	2021
Road network	OpenStreetMap (openstreetmap.org)	–	2021
Land cover	GlobeLand30 (Chen et al., 2015)	30 m	2020
Total population and age structure	WorldPop (WorldPop, 2018)	100 m	2020
GDP	Statistical yearbooks	–	2020

2.2. Data sources and data preprocessing

The data employed in this paper included POIs of schools and hospitals, the OSM road network, the land cover of China, and spatial demographic data from WorldPop. The data sources and descriptions are shown in Table 1.

The POI of schools and hospitals in this study was obtained based on the application program interface of the Gaode map. Based on the classification code of Gaode map POI, we first selected two major categories, namely, *science/cultural & education service* and *medical service*, which correspond to the distribution of schools and hospitals. In this study, the category of *science/cultural & education service* includes three subcategories: kindergarten, elementary school, and middle school; the medical service includes the subcategories such as hospital, special hospital, clinic, health center, and pharmacy.

The road network data used in this study were obtained from the historical road network data of OSM in 2021. In this study, we selected the fields of “railway” and “road” to construct a road network in China. In the field “road”, we selected five types: motorway (motorway link), trunk (trunk link), primary (primary link), secondary (secondary link),

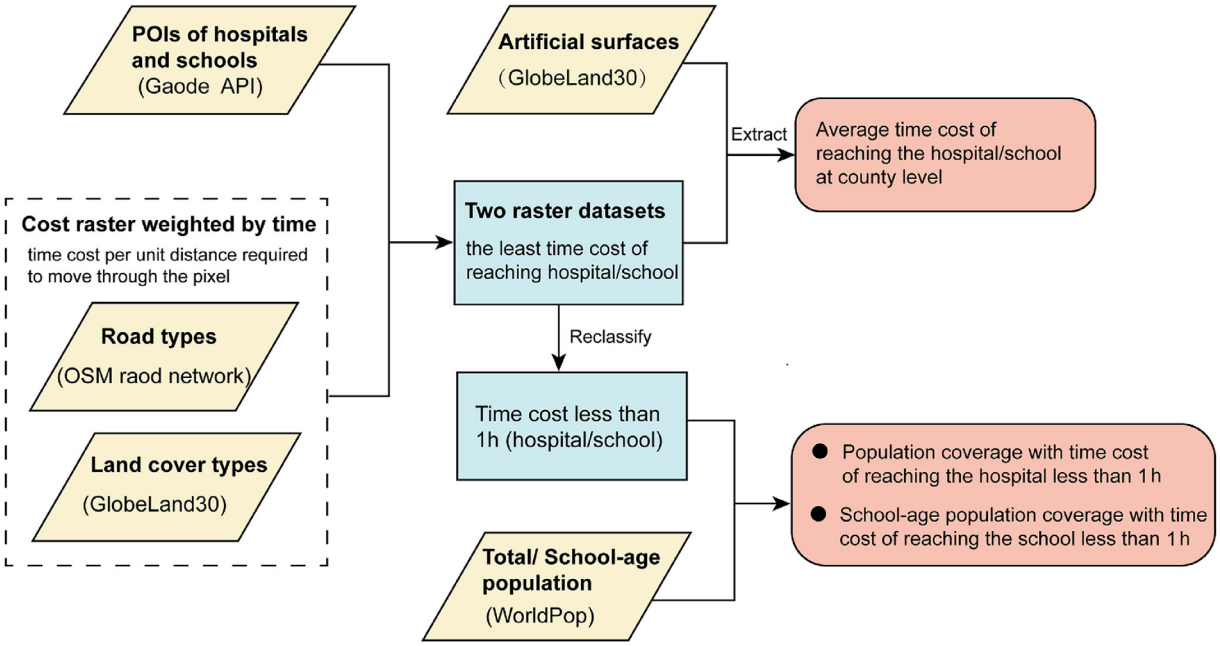

Fig. 2. Calculation of accessibility metrics of health and education services.

and tertiary. We excluded the detailed road features below the "tertiary". After comparing the OSM road network data with both the Gaode and Baidu Street maps of China, we found that the "railway" and the "road" features retained in our study have an acceptable accuracy in position and completeness.

The land cover data used in this study were the public global land cover dataset: GlobeLand30 (Chen et al., 2015). The dataset was produced based on Landsat satellite (TM/ETM+) images and Chinese Environmental and Disaster satellite (HJ-1) images and includes 11 surface land cover types: cultivated land, forest, grassland, shrub, wetland, water body, tundra, artificial surfaces, bare land, glacier, and permanent snow. The land cover data in 2020 will be used to generate a time cost raster combined with the road network.

The population count and age and gender structure data of China in 2020 were obtained from WorldPop spatial demographic data (WorldPop, 2018). The population count data provide the estimated total number of people per grid cell (100 m). The age and gender datasets provide an estimated total number of people per grid (100 m) broken down by gender and age groupings.

2.3. Calculation of accessibility metrics of health and education services

The accessibility metrics of health and education services include the least time cost to reach hospitals and schools and the population coverage with time cost of less than 1 h. The calculation process is shown in Fig. 2.

The POI of schools and hospitals, OSM road network, and land cover of China were comprehensively used to calculate the least time cost of reaching the hospital and school. We used the cost-distance tool in ArcMap software to calculate the time cost. The tool can calculate the nearest weighted distance from each pixel to the input source data over a cost raster. Here, we took the calculation of the least time cost of reaching the hospital as an example to explain the calculation process. The POIs of the hospitals were treated as the source data. Then, a cost raster was generated by combining the OSM road network data and the land cover. The time cost weight of each pixel was assigned according to its road type or land cover type, as shown in Tables 2 and 3. The time cost settings of road and land cover types refer to the weight settings in the analysis of global travel accessibility assessed by Weiss et al. (2018). The weight allocated to each pixel represents the time cost per unit of distance required to move through the pixel (Hu et al., 2022). The least time cost of reaching the hospital of each pixel was calculated by multiplying the pixel size by the time cost value. The least time cost of reaching the school was then calculated through the same calculation process. Finally, two raster datasets of the least time cost of reaching the hospital and the school with 30 m spatial resolution were generated to evaluate the access to health and education services in China in 2021. Further, to measure the mean time cost at the county level, we extracted the value where the urban area was located using artificial surfaces land cover type in GlobeLand30 and then averaged the time cost value to the county level.

Table 2
Time cost weight of each road type.

Name	Time cost of per unit moving (min/km)
Railway	0.5
Motorway; motorway link	0.5
Trunk; trunk link	1
Primary; primary link	1
Secondary; secondary link	1.5
Tertiary; tertiary link	6

Based on the two generated time cost raster datasets, pixels with a time cost value less than 1 h were both extracted. Combining the WorldPop spatial demographic data, the population coverage with the time cost of reaching the hospital in less than 1 h (H_{1h}) was then calculated using Eq. (1). The WorldPop spatial demographic data provides the age

Table 3
Time cost weight of each land cover type.

Land cover type	Time cost of per unit moving (min/km)
Cultivated land	36
Forest	48
Grass land	36
Shrub land	36
Wet land	48
Water	3
Tundra	48
Artificial land	6
Bare land	24
Glaciers and permanent snow	48

structure data as age groups per five years, and groups 5–9, 10–14, and 15–19 were selected in this study. According to China's average enrollment age of kindergarten and graduate age of high school, the population aged 5–19 basically covered the students in kindergarten, elementary school, and middle school (junior school and high school) in China (Hu et al., 2010; Hu, 2012; Chen, 2015). Therefore, in this study, we treated the population aged 5–19 years as the school-age population. Combined with the school-age population, the school-age population coverage with the time cost of reaching the school less than 1 h (S_{1h}) was then calculated using Eq. (2).

$$H_{1h} = \frac{\text{Population with time cost} \leq 1\text{h}}{\text{Total population}} \times 100\% \tag{1}$$

$$S_{1h} = \frac{\text{school age population with time cost} \leq 1\text{h}}{\text{school age population}} \times 100\% \tag{2}$$

2.4. Cluster map of county-level time costs

The local indicators of spatial association (LISA) index was used to generate the cluster map of county-level mean time cost (Anselin, 1995). Local Moran's I was used to obtain cluster maps of the LISA in this study, including statistically significant clusters of high values (high–high), and clusters of low values (low–low), as taken from GeoDa software (Anselin et al., 2010). Taking the least time cost to hospital as an example, local Moran's I is given as follows:

$$I_i = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij} \times (x_i - \bar{x})(x_j - \bar{x})}{\frac{1}{n}(x_i - \bar{x})^2} \tag{3}$$

where, I_i is local Moran's I for county i, x_i is the least time to reach hospital for county i, $\bar{x}$ is the mean value of the corresponding time cost, w_{ij} is the spatial weight between counties i and j, and n is the total number of counties.

2.5. Z-score

The Z-score was used in this study to normalize the time cost value for each county. The Z-score describes the relationship between a given value and the mean value, which is measured in terms of the standard deviations from the mean. The Z-score is given as follows:

$$Z = \frac{X - \bar{X}}{S} \tag{4}$$

where, X is the time cost for a certain county, $\bar{X}$ is the mean value of the time cost, and S is the standard deviation of the time cost. Z-scores may be positive or negative, with a positive value indicating that the score is above the mean and a negative score indicating that it is below the mean. Furthermore, a Z-score equal to 0 means that the given value is identical to the mean value, and a Z-score equal to 1 means that the given value is one standard deviation from the mean. Given that a higher time cost indicates lower accessibility to public services, we took the opposite value of the Z-score to better understand the relationship between a given time cost and the average level in this study.

3. Results

3.1. Spatial distributions and statistical characteristics of accessibility metrics

For the county-level mean time cost, as shown in Fig. 3(a) and (b), the spatial distribution of the time cost of reaching the hospital and the time cost of reaching the school in China presented a similar pattern, which was generally low in the east and high in the west. The cluster map based on LISA clearly identified this spatial pattern, as illustrated in Fig. 3(c) and (d). Northwest China, Tibet, Inner Mongolia, and Northern Heilongjiang Province showed statistically significant clusters of high time costs, indicating low accessibility to both hospitals and schools. The counties in Central China, East China, and South China showed statistically significant clusters of low time cost, indicating high accessibility to hospitals and schools. The time cost of reaching the hospital and the school of all counties in these regions was less than 30 min, whereas the counties with the time cost greater than 2 h were both located in Northwest and Tibet. The counties with low accessibility to hospitals and schools were mainly located in Tibet and Qinghai Province, whereas the counties with high accessibility were located in the eastern seaboard, such as Shandong Province, Jiangsu Province, and Zhejiang Province.

For the county-level population coverage, Fig. 3(e) and (f) show the spatial distribution of population coverage with the time cost of reaching the hospital less than 1 h and school-age population coverage with the time cost of reaching the school less than 1 h, respectively. The overall

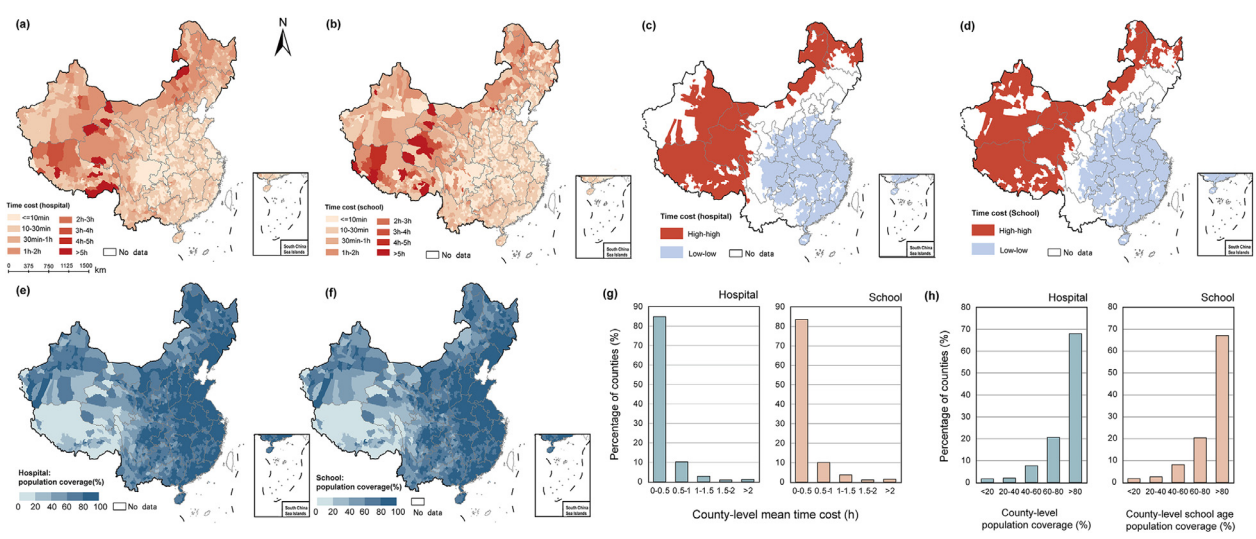

Fig. 3. Spatial distributions and statistical characteristics of the accessibility metrics. (a) The time cost of reaching the hospital; (b) the time cost of reaching the school; (c) cluster map for time cost of reaching the hospital; (d) cluster map for time cost of reaching the school; (e) population coverage with the time cost of reaching the hospital less than 1 h; (f) school-age population coverage with the time cost of reaching the school less than 1 h; (g) histograms of the time cost to reach the hospital and school; (h) histograms of the population coverage of the time cost to reach the hospital less than 1 h and school-age population coverage of the time cost to reach the school less than 1 h.

pattern both presented a high accessibility in the east and a low accessibility in the west. The counties with a population coverage of less than 20% were mainly located in Tibet, whereas those with a population coverage greater than 80% were located in the central and eastern regions.

For the time cost, on the basis of land surface information, the overall county-level mean time cost to reach the hospital and school was 20 and 22 min, respectively. The histograms of the time cost to reach the hospital and school are plotted in Fig. 3(g), where the horizontal axis is the county-level mean time cost and the vertical axis is the percentage of counties. In 95.1% of counties in China, the average time cost of reaching the hospital was less than 1 h. Only 1.1% of counties had an average time cost longer than 2 h. In more than 93.7% of counties in China, the average time cost of reaching the school was less than 1 h, whereas 1.1% of counties took longer than 2 h.

For population coverage, on the national scale, 94.7% of the population in China had a time cost of reaching the hospital less than 1 h. The histograms of the county-level population coverage are plotted in Fig. 3(h), where the horizontal axis is the county-level population coverage with a time cost of reaching the hospital less than 1 h and school-age population with a time cost of reaching the school less than 1 h, respectively, and the vertical axis is the percentage of counties. At the county level, 68% of the counties had population coverage exceeding 80%, whereas 11.4% of the counties had less than 60% (Fig. 3(h)). On the national scale, 92.5% of the school-age population in China had a time cost of reaching the school less than 1 h. At the county level, 67% of the counties had a school-age population coverage of time cost to a school less than 1 h exceeding 80%, whereas 12.5% of the counties had less than 60% (Fig. 3(h)).

3.2. Regional disparities in accessibility of health and education services

3.2.1. Regional disparities in geographical divisions

In order to analyze the regional disparities in the accessibility of hospitals and schools, we focused on three groups of areas which include counties in different geographical divisions, non-poor counties vs. poor counties, and ecological function counties vs. non-ecological counties, as shown in Fig. 4. For the regional disparities among different geographical divisions, as shown in Fig. 4(a), Central China, East China, and South China had high accessibility to both hospitals and schools (low time cost of reaching both hospitals and schools). The time cost in more than half of the counties was less than 15 min in the three regions. Furthermore, the range of interquartile in the three regions was narrow, indicating the relatively balanced time cost within each region. The Northeast region both have higher time costs for reaching the hospital and school compared with other regions, and their wide range of interquartile indicates an unbalanced time cost within the region. Followed is the Northwest, North China, and Southwest regions which also have high time costs of reaching the hospital and school. In the Southwest region, the mean value was much higher than the median value. This is mainly because the time cost was very high in areas such as Tibet and Qinghai, and raised the mean value of the entire region. However, the accessibility of hospitals and schools in most counties in the Southwest was not that low, and the time cost in half of the counties in the Southwest was less than 30 min. North China presented a similar situation to the Southwest, and the time cost of the counties in Inner Mongolia was much higher than the other provinces in North China.

3.2.2. Regional disparities in poor areas and ecological function areas

Boxplots of the time cost of reaching the hospital and school in poor areas and non-poor areas are presented in Fig. 4(b). The time cost in poor areas was obviously higher in the upper limit, mean, and median values than in the non-poor areas, indicating the quite lower accessibility to hospitals and schools in poor areas. As shown in Table 4, the mean time cost in poor areas was more than twice that in non-poor areas. The median time cost was also much higher in poor areas than in non-poor areas. The time costs of the ecological function counties and the non-ecological function counties are plotted in Fig. 4(c). Similar to the poor areas versus non-poor areas, the statistical results showed that the time cost of the ecological function counties was higher than the non-ecological function counties in terms of the upper limit, mean, and median values. The mean and median time cost in ecological function counties was approximately twice that in non-ecological function counties (Table 4).

Table 4
Mean and median time cost in poor versus non-poor and ecological versus non-ecological areas (unit: h).

	Poor	Non-poor	Ecology	Non-ecology
Hospital (mean)	0.60	0.23	0.69	0.23
School (mean)	0.61	0.28	0.63	0.30
Hospital (median)	0.30	0.14	0.34	0.15
School (median)	0.27	0.17	0.29	0.17

In order to better quantify the inequality distribution of health and education services, we plotted Fig. 4(d) with reference to the Loren curve (Lorenz, 1905). Lorenz curve plot the cumulative share of income from different sections of the population (Gastwirth, 1972). Similarly, Fig. 4(d) plots the relationship between the ranking percentile of China's county-level GDP and the corresponding cumulative time cost percentile. The horizontal axis represents the ranking percentile of county-level GDP (the county-level GDP ranked from low to high), and the vertical axis is the cumulative percentile of the corresponding time cost. The greater the curvature, the higher the degree of inequality. The apparent curvature indicates that counties with low accessibility to hospitals and schools (high time cost) tend to concentrate in counties with low GDP than in counties with high GDP. As shown in Fig. 4(d), the intersection points show that the cumulative time cost of hospitals and schools in the bottom 20% of counties ranked by GDP reached approximately 80% of the national total. This indicates an uneven distribution of health and education resources, meaning that low-GDP counties require disproportionately increased time costs to reach medical resources compared with high-GDP counties. Furthermore, a more severe uneven was observed in the allocation of education resources, as shown in the higher curvature of the curve than that of the hospital Fig. 4(d).

3.3. Uneven distribution of schools and hospitals within each county

On the basis of the *Z*-score, as shown in Fig. 5, the counties were first plotted into four quadrants, where quadrant I indicates the counties with high accessibility to both hospital and school (both the time costs of reaching the hospital and school are lower than the national average level), quadrant II indicates low accessibility to hospital but high accessibility to school, quadrant III indicates low accessibility to both hospital and school, quadrant IV indicates high accessibility to hospital but low accessibility to school.

We focused on the counties located in quadrants II, III, and IV. Then, the counties in quadrants II, III, and IV were further classified into six categories using the 80% quantile (Fig. 5(a)). In quadrant II, the counties were further divided into two categories: (1) slightly low accessibility to hospital, with the time cost of reaching the hospital higher than the average but lower than the 80% quantile (5.7% of counties); and (2) significantly low accessibility to hospital, with time costs higher than the 80% quantile (6.8% of counties). In quadrant III, the counties are further divided into two categories: (1) slightly low accessibility, that is, the time costs of reaching the hospital and school were both higher than the average but lower than the 80% quantile (4.2% of counties); and (2) significantly low accessibility, with time costs higher than the 80% quantile (15.6% of counties). In quadrant IV, the counties were further divided into two categories: (1) slightly low accessibility to school, that is, the time cost of reaching the school was higher than the average but lower than the 80% quantile (3.9% of counties); and (2) significantly

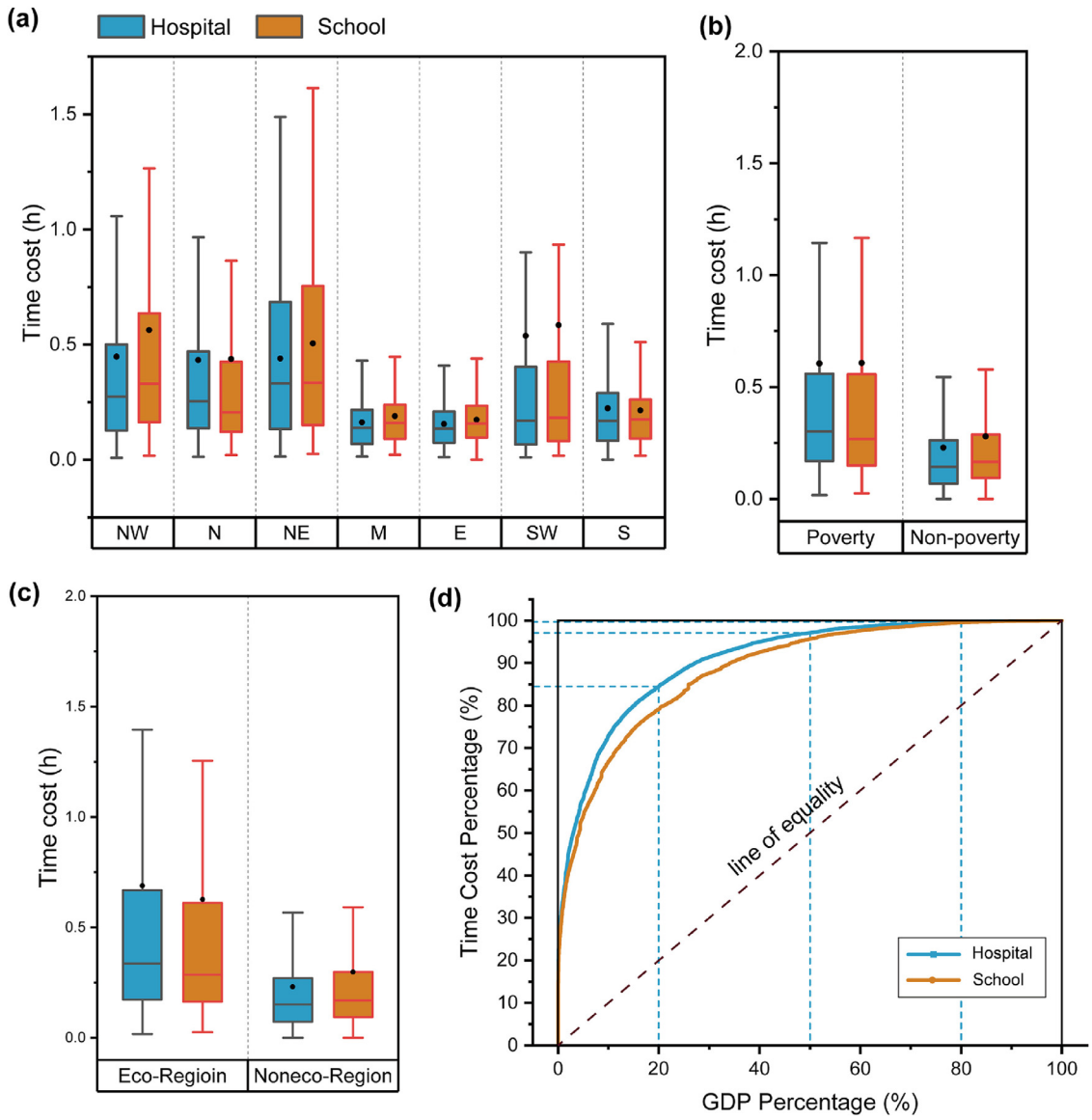

Fig. 4. Regional disparities in accessibility of health and education services. (a) Different geographical divisions; (b) poor vs. non-poor areas; and (c) ecological vs. non-ecological function areas. (d) Inequality curves of the time cost for hospitals and schools, shown as a function between the ranking percentile of China's county-level GDP and the percentile of accumulative time cost for corresponding counties. Northeast: NE; East China: E; North China: N; Central China: M; South China: S; Southwest: SW; Northwest: NW.

low accessibility to school, with time costs higher than the 80% quantile (6.9% of counties).

As shown in Fig. 5(b), the counties located in quadrant III which have low accessibility to both hospitals and schools tend to cluster in the west of China. The counties with significantly low accessibility to both hospitals and schools show scattered distribution. The categories of significantly low accessibility to hospital and significantly low accessibility to school in quadrants II and VI were also scattered in the whole country. Meanwhile, they are inclined to concentrate in the Northeast region, Tibet and Yunnan Province. The allocation of schools and hospitals was quite uneven within these counties.

4. Discussion

4.1. Uneven distribution of health and education services

Overall, the results revealed an evident spatial pattern with "high accessibility to hospital and school in the east and low accessibility in the west". This pattern is consistent with the disparities in economic growth and population density between the west and east of China, which are both high in the east and low in the west, as shown in Fig.1. Counties with low accessibility to health and education services are highly coupled with impoverished counties and ecological function counties. The results indicate that the time costs in the poor areas and ecological areas are approximately doubled than in the non-poor and non-ecological areas. The relative backward health and education services in poor areas are usually highly related to their geographical characteristics, given that the poor areas in China are mostly distributed in the mountain areas, arid and semi-arid areas, and border areas (Chen and Ge, 2015). The harsh and closed geographical conditions restrain socioeconomic development and slow the infrastructure construction process in poor areas, resulting in low accessibility to public services. Fortunately, over the past two decades, China has made great efforts to improve the infrastructure in poverty-stricken areas; an example is the *entire-village advancement programs*. As for the ecological areas, as shown in Fig. 1, large overlapping areas occur between poor and ecological

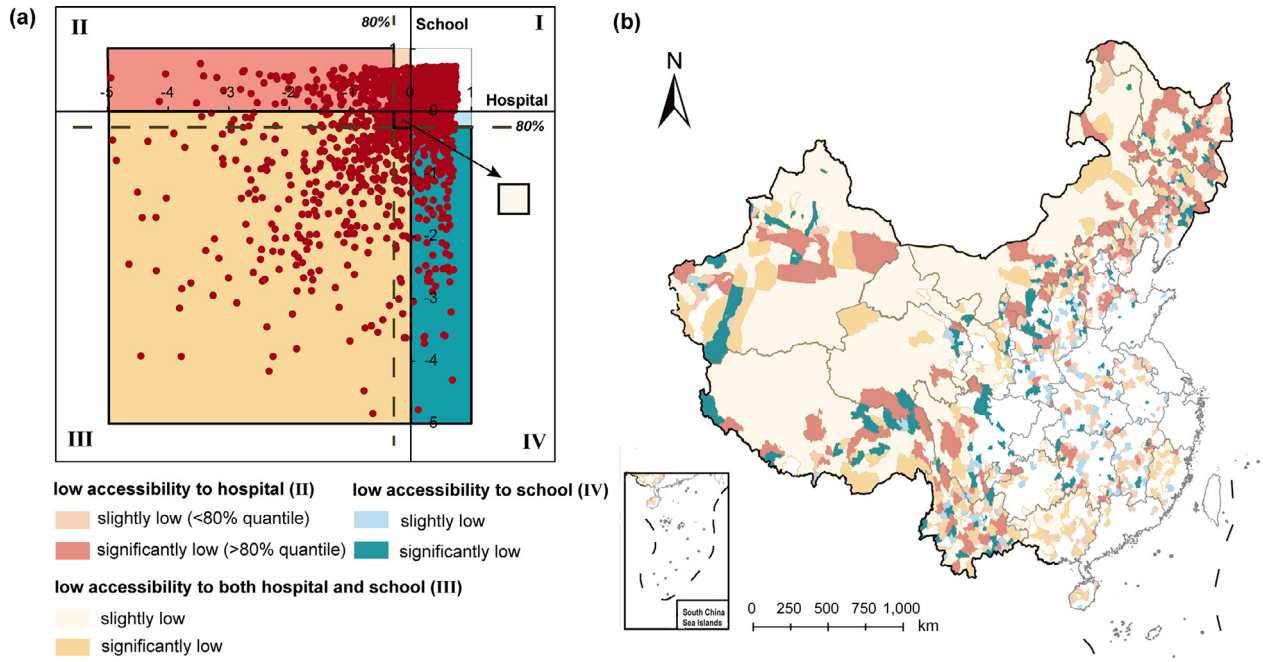

Fig. 5. Uneven distribution of hospitals and schools within each county. (a) Quadrantal distribution of the *Z*-score for all counties excluding the points with a time cost of more than 5 h. (b) Spatial distribution of the *Z*-score as classified by the 80% quantile.

areas. Therefore, natural conditions are also key factors resulting in low accessibility to health and education services in ecological function areas. The ecological function counties mainly included ecologically fragile areas with harsh natural environments and protected areas that shoulder ecological construction responsibilities. Infrastructure construction is difficult to promote in areas with harsh natural environments, and protected areas usually strictly restrict excessive development. Consequently, accessibility to public services in ecological areas is relatively backward compared with non-ecological areas. China has also addressed this problem in the process of poverty reduction; for instance, the *resettlement program* has been implemented in ecologically fragile areas since 2000 and has improved living standards and access to health, education, and utilities by relocating impoverished populations from living unfriendly areas (Yang et al., 2020). The accessibility of health and education resources has been continuously improved in the west of China in recent years, particularly in poor areas (SCIO, 2021). However, the inequality distribution of public services between western and eastern China, poor and non-poor areas is hard to eliminate in a short time, because it is also a complex systematical issue of regional development. But the targeting policy interventions can narrow the inequality. Strengthening the infrastructure construction in the counties with extremely low accessibility is still the key to minimizing the time cost to reach health and education services. Besides, the proper planning and allocation of health and education services are also critical to improve accessibility.

In general, the time cost of reaching the hospital and school presented a similar pattern; most of the counties have low or high accessibility to both hospitals and schools at the same time. However, this study also identified some counties with uneven distribution of hospitals and schools within the counties. These counties could be divided into two categories: (1) high accessibility to a hospital but low accessibility to a school, and (2) high accessibility to a school but low accessibility to a hospital. These counties are mainly distributed in the areas from the three provinces in the Northeast to Hebei, then to Shanxi, and finally to Yunnan Province (Fig. 5(b)). Apart from the counties with low accessibility to both hospitals and schools, the uneven allocation between hospitals and schools within counties also needs to be further addressed.

4.2. Limitations

This study has certain limitations that need to be addressed. Using multi-source geospatial data allows the evaluation of some socioeconomic outcomes in a cost-effective way. However, the uncertainty of geospatial data cannot be ignored. In such a large-scale evaluation, the data uncertainty introduced by the excessive amount of POI and OSM road network data is unpredictable, such as the position and attribution accuracy of each POI point and road, as well as the uncertainty in data redundancy and incompleteness (Themistocleous et al., 2016). Furthermore, in the process of calculating the time cost, we considered land surface information, which is land cover and road networks, but not other barriers, such as traffic congestion, ways of traveling, and human behavior-related information. In addition, the information on supply and demand also was not included. Consequently, the time cost value calculated in this paper is more of a relative physical travel time that measures the accessibility of health and education services on the basis of transportation information and land cover. In future work, we hope that the supply and demand of public services and human-related information could be added to improve the accuracy of accessibility metrics.

5. Conclusions

This study proposed two accessibility metrics that evaluate access to the hospital and school using open geospatial data and a satellite imagery-based data product, which include POI, OSM road network, WorldPop spatial demographical data, and land cover. The results show that (1) on the basis of the land surface information, the overall average time cost of reaching the hospital and school is 20 and 22 min across the country, respectively. On the national scale, 94.7% of the population in China has a time cost of reaching the hospital less than 1 h, and 92.5% of the school-age population has a time cost of reaching the school less than 1 h. (2) The spatial distribution presents an evident pattern of “high accessibility to hospitals and schools in the east and low accessibility in the west”. The counties with low accessibility are highly coupled with poor areas and ecological function areas, and the time costs in the poor and ecological areas are more than twice in the non-poor and non-ecological areas. (3) The cumulative time cost of hospital

and school in the bottom 20% of counties (ranked by GDP) reached approximately 80% of the national total, indicating that low-GDP counties require disproportionately increased time costs to reach public services compared with high-GDP counties. (4) The counties with the uneven allocation of hospitals and schools are mainly distributed from the three provinces in the northeast to Hebei, and to Shanxi, and finally to Yunnan Province. These findings quantify the disparities between the west and the east, poor and non-poor areas in the accessibility of health and education services, which can help identify areas with backward public services and may contribute to targeted and efficient policy interventions. Furthermore, the proposed accessibility metrics in this study can be a type of auxiliary data to better help evaluate SDG outcomes.

Declaration of Competing Interests

The authors declare that there are no known competing financial interests or personal relationships that influenced the work reported in this paper.

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